Day 31

The Promise API:

There are two types of methods:

A) Instance Methods

(Methods used *on a promise you created*)

Instance Methods

- **then()** → success
- **catch()** → error
- **finally()** → always

B) Static Methods

(Methods used *on the Promise class itself*, like Promise.all())

Static Methods

- **Promise.resolve()** → make resolved promise
- **Promise.reject()** → make rejected promise
- **Promise.all()** → wait for all (fails if one fails)
- **Promise.allSettled()** → wait for all (never fails)
- **Promise.race()** → first finished
- **Promise.any()** → first successful

Example: basic example to print the resolved value of the promises in the console.

Here, after 1s, Value 1 is printed, after 2s Value 2 is printed and after 3s, Value 3 is printed.

Example: doing the same thing we did above using the promise.all()

```
30  let p1 = new Promise((resolve, reject) => {
31      setTimeout(() => {
32          resolve("Value 1");
33      }, 1000);
34  });
35
36  let p2 = new Promise((resolve, reject) => {
37      setTimeout(() => {
38          resolve("Value 2");
39      }, 2000);
40  });
41
42  let p3 = new Promise((resolve, reject) => {
43      setTimeout(() => {
44          resolve("Value 3");
45      }, 3000);
46  });
47
48  let promise_all = Promise.all([p1,p2,p3]);
49  promise_all.then((values) => {
50      console.log(values);
51  });
```

```
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[ 'Value 1', 'Value 2', 'Value 3' ]
```

But what if one of the promise get rejected? Will promise.all() still works? No!

```
54  let p1 = new Promise((resolve, reject) => {
55      setTimeout(() => {
56          reject("Some error occurred in p1");
57      }, 1000);
58  });
59
60  let p2 = new Promise((resolve, reject) => {
61      setTimeout(() => {
62          resolve("Value 2");
63      }, 2000);
64  });
65
66  let p3 = new Promise((resolve, reject) => {
67      setTimeout(() => {
68          resolve("Value 3");
69      }, 3000);
70  });
71
72  let promise_all = Promise.all([p1,p2,p3]);
73  promise_all.then((values) => {
74      console.log(values);
75  });
```

```
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PS E:\JavaScript\Day31-40\Day31> node main.js
      new UnhandledPromiseRejection(reason);
      ^

UnhandledPromiseRejection: This error originated either by throwing inside of an async function without a catch block, or by rejecting a promis
(). The promise rejected with the reason "Some error occurred in p1".
```

How to deal with this? We can use .catch() or can use Promise.allSettled() as shown below:

```
77  let p1 = new Promise((resolve, reject) => {
78      setTimeout(() => {
79          reject("Some error occurred in p1");
80      }, 1000);
81  });
82
83  let p2 = new Promise((resolve, reject) => {
84      setTimeout(() => {
85          resolve("Value 2");
86      }, 2000);
87  });
88
89  let p3 = new Promise((resolve, reject) => {
90      setTimeout(() => {
91          resolve("Value 3");
92      }, 3000);
93  });
94
95  let promise_all = Promise.allSettled([p1,p2,p3]);
96  promise_all.then((values) => {
97      console.log(values);
98  });
```

```
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PS E:\JavaScript\Day31-40\Day31> node main.js
[
  { status: 'rejected', reason: 'Some error occurred in p1' },
  { status: 'fulfilled', value: 'Value 2' },
  { status: 'fulfilled', value: 'Value 3' }
]
```

Now, what if we wish that the promise which first get resolved will get printed? Then we can use Promise.race():

```
100 let p1 = new Promise((resolve, reject) => {
101     setTimeout(() => {
102         resolve("Value 1");
103     }, 111000);
104 });
105
106 let p2 = new Promise((resolve, reject) => {
107     setTimeout(() => {
108         resolve("Value 2");
109     }, 2000);
110 });
111
112 let p3 = new Promise((resolve, reject) => {
113     setTimeout(() => {
114         resolve("Value 3");
115     }, 3000);
116 });
117
118 let promise_all = Promise.race([p1,p2,p3]);
119 promise_all.then((values) => {
120     console.log(values);
121 });
```

```
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PS E:\JavaScript\Day31-40\Day31> node main.js
Value 2
```
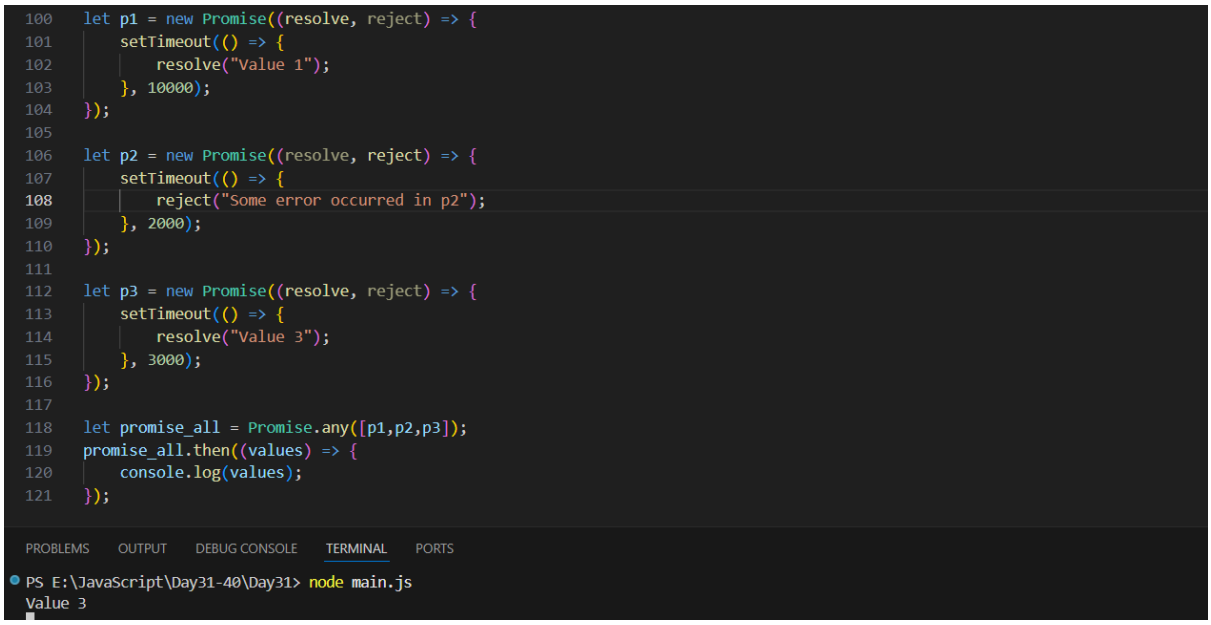

What if the fastest one giving an error? Then .race() will show error. How to deal then? We can use .any():



```
let p1 = new Promise((resolve, reject) => {
    setTimeout(() => {
        resolve("Value 1");
    }, 10000);
});

let p2 = new Promise((resolve, reject) => {
    setTimeout(() => {
        reject("Some error occurred in p2");
    }, 2000);
});

let p3 = new Promise((resolve, reject) => {
    setTimeout(() => {
        resolve("Value 3");
    }, 3000);
});

let promise_all = Promise.any([p1,p2,p3]);
promise_all.then((values) => {
    console.log(values);
});
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS E:\JavaScript\Day31-40\Day31> node main.js
Value 3
```

--The End--